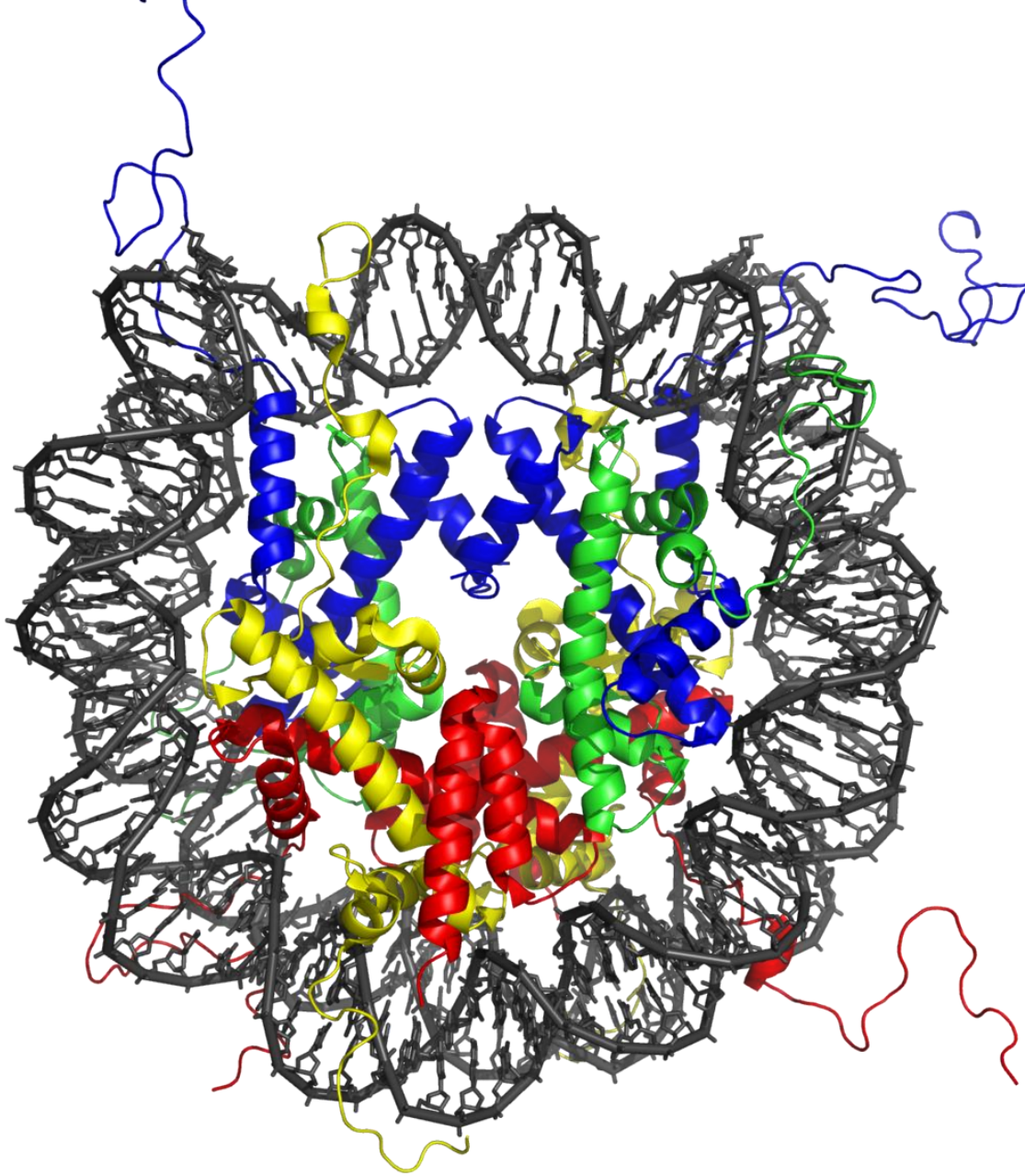
DP Biology – A1.2 (HL extension)

# DNA: direction, pairing and packing

- How do nucleic acids store information that can be inherited?
- Which features of DNA allow it to be copied accurately?

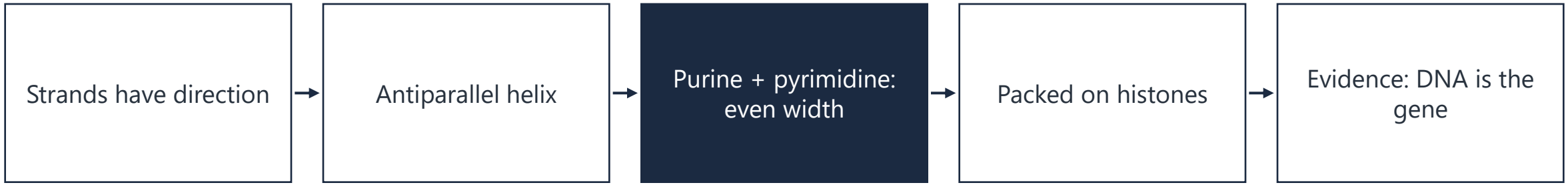


HL statements A1.2.11–15. Textbook pp. 25–33.
Image: a nucleosome, DNA wrapped around eight histones. Zephyris, CC BY-SA 3.0.

# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| **1** | A1.2.11–12 | Explain the 5′ to 3′ direction of strands, and why purine-pyrimidine pairing keeps the helix regular |
| **2** | A1.2.13 | Describe the structure of a nucleosome, and explain why DNA is packed this way |
| **3** | A1.2.14–15 | Evaluate the evidence from Hershey and Chase, and from Chargaff's data, for the genetic material |

# The big picture: a molecule built to be copied



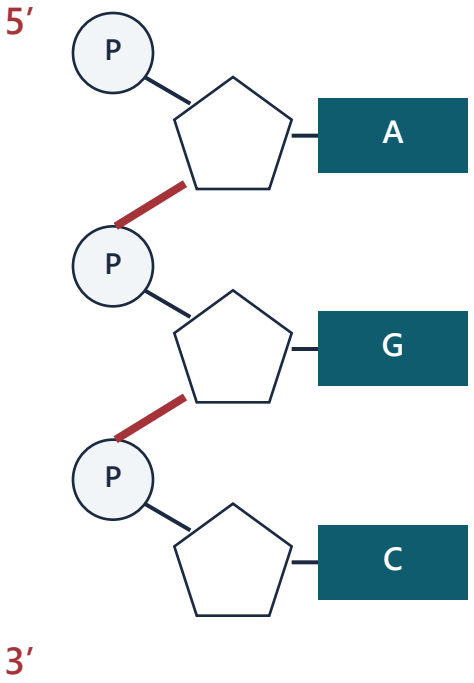

DNA's **structure** explains how it is copied and read; classic **experiments** showed it, not protein, carries genes.

**Big question**

How did scientists prove that genes are DNA, when most expected protein?

# Every strand has a direction

A1.2.11




Phosphate links carbon 3′ to carbon 5′

Sugar carbons are numbered 1′ to 5′. Each phosphate joins **carbon 3′** of one sugar to **carbon 5′** of the next.

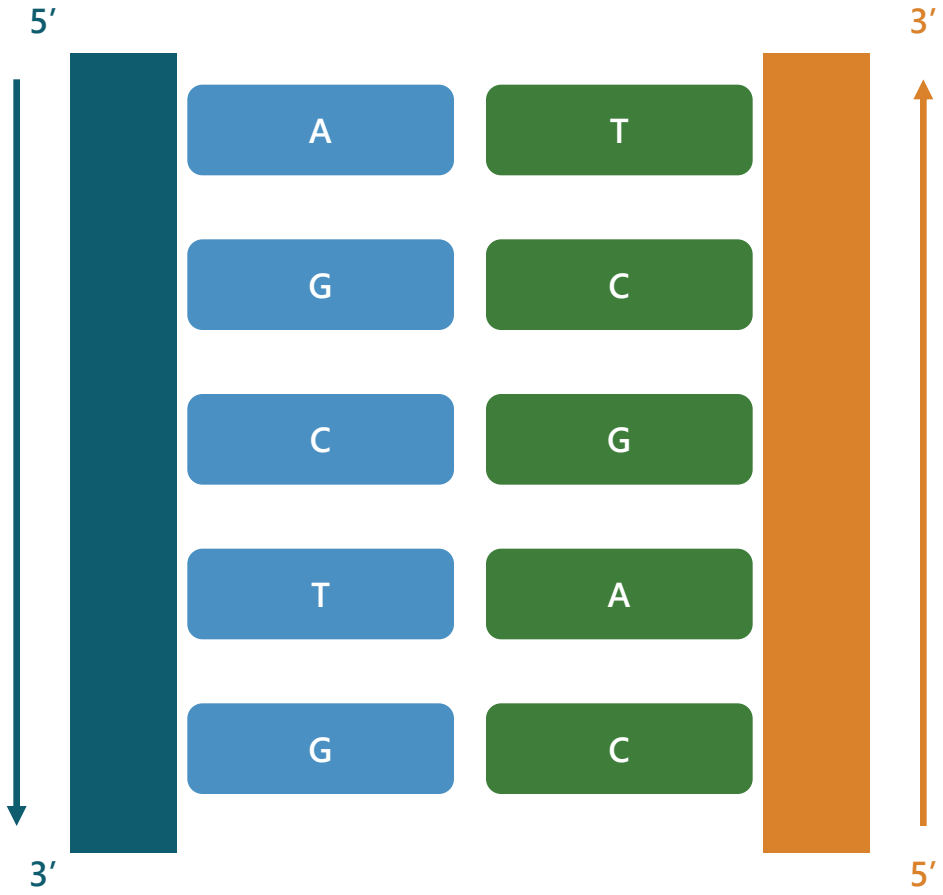
So a strand has a **5′ end** (free phosphate) and a **3′ end** (free -OH).

## Key idea

Nucleotides are only added to the free 3′ -OH, so strands are always built 5′ to 3′.

# Antiparallel strands






The two strands run in **opposite directions**: they are **antiparallel**.

Bases pair across the middle: A with T, G with C.

# Why direction matters

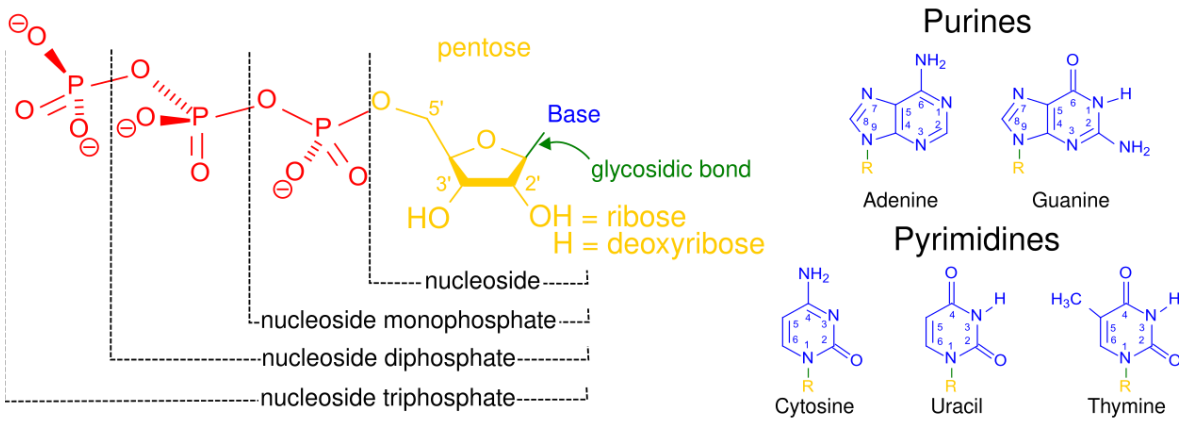
A1.2.11

| Process | Role of direction |
| --- | --- |
| **Replication** | New strands grow 5′ to 3′; the antiparallel strands are copied in opposite directions |
| **Transcription** | The template is read 3′ to 5′; mRNA is built 5′ to 3′ |
| **Translation** | Ribosomes read mRNA from the 5′ end |

**Link**
D1.1 and D1.2 (HL): leading and lagging strands; transcription and translation.

# Purines pair with pyrimidines

A1.2.12




Nucleotide parts, purines and pyrimidines. Public domain.

| | Purines | Pyrimidines |
|---|---|---|
| **Rings** | Two | One |
| **Bases** | A, G | C, T (U in RNA) |

Every base pair is **one purine + one pyrimidine**, so all pairs are the **same width** and the helix is even.

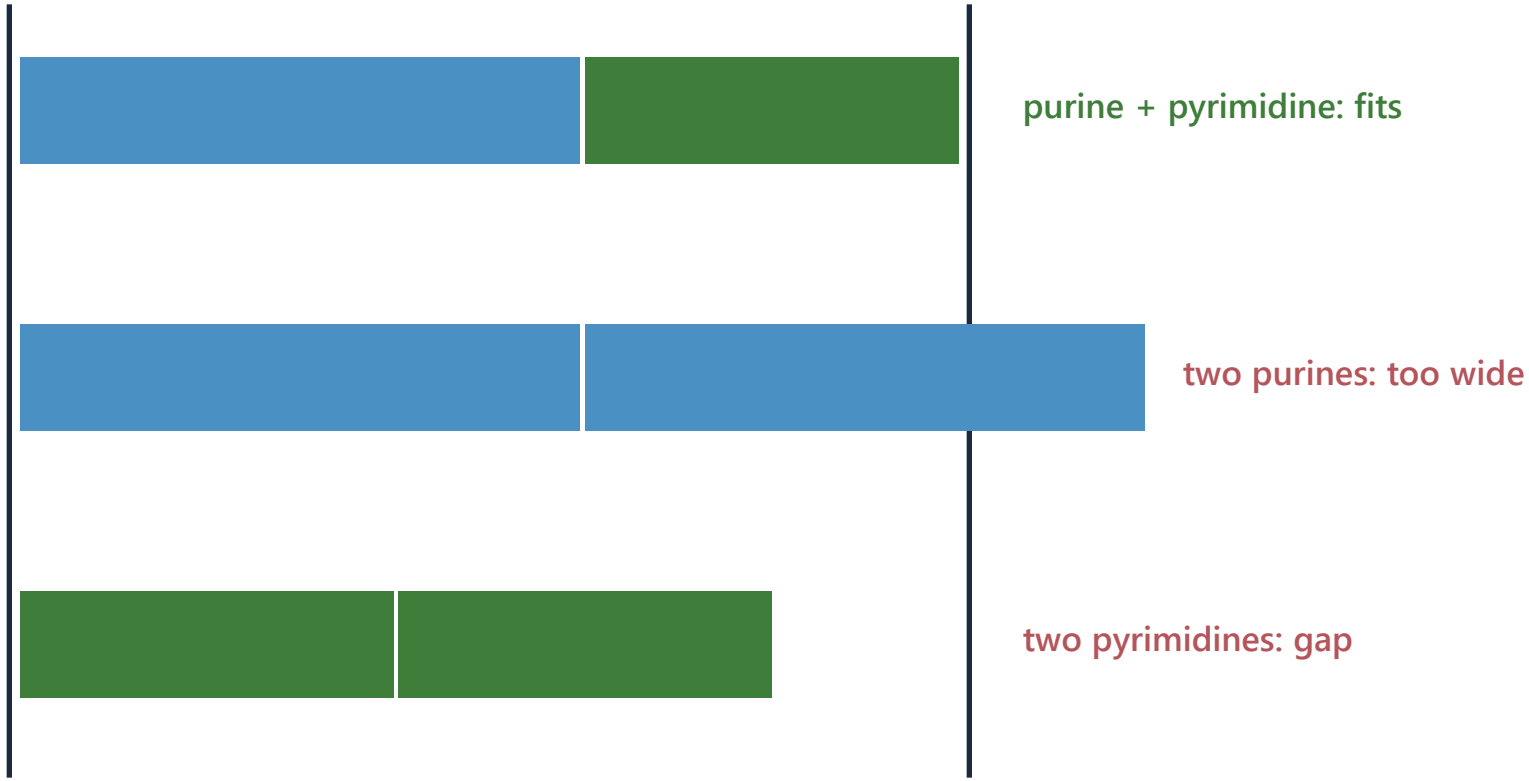
A-T has **two** hydrogen bonds; C-G has **three**.

**So**

Any sequence can be stored without changing the molecule's shape.

# Why only purine + pyrimidine fits

A1.2.12




**Key idea**

Two rings + one ring always gives the same width.

# Working with data: C-G pairs and stability

A1.2.12

Melting temperature: half the DNA strands separated. Values from the published Marmur-Doty relationship (1962), rounded.

| C-G pairs / % | Melting / °C |
|---|---|
| **30** | 82 |
| **40** | 86 |
| **50** | 90 |
| **60** | 94 |
| **70** | 98 |

1. Describe the relationship.
2. Explain it using hydrogen bonds.
3. Predict the value for 45% C-G.
4. Suggest why hot-spring bacteria often have high C-G.

# Check-in: class 1

A1.2.11–12

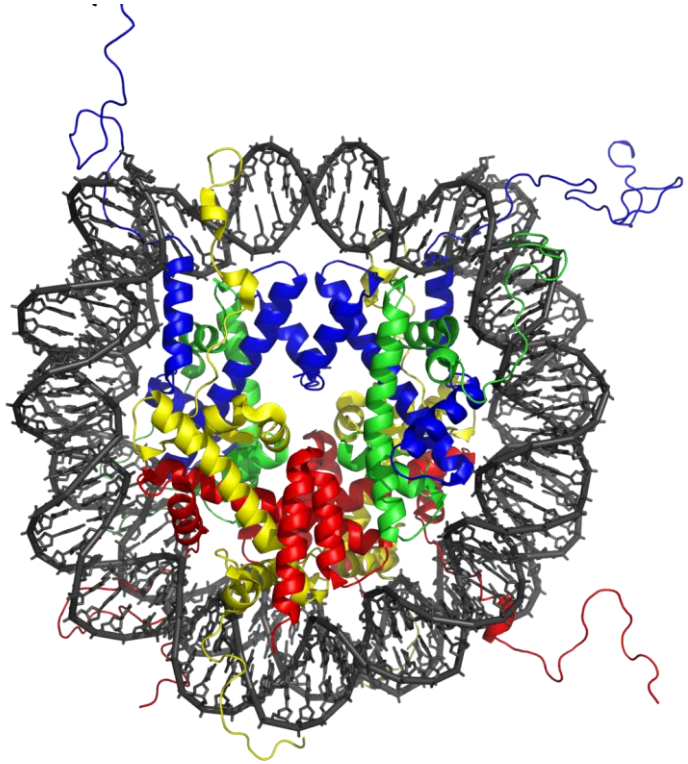
1. State which end new nucleotides are added to.
2. Explain what antiparallel means.
3. State the number of rings in purines and pyrimidines.
4. Explain why purine-pyrimidine pairing keeps the helix regular.
5. Explain why C-G-rich DNA melts at a higher temperature.

**How to use this**

Answer without your notes, then check against slides 3–9.

# The nucleosome





Eukaryotic DNA winds about twice around a core of **eight histones** (two each of four types): a **nucleosome**.

**Linker DNA** joins nucleosomes; histone **H1** holds the DNA where it enters and leaves.

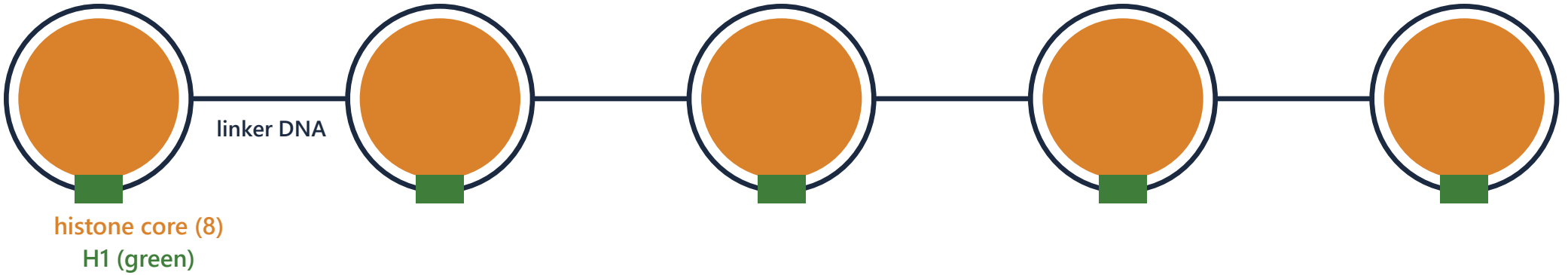
Positive histones (lysine, arginine) attract negative DNA phosphates.

Nucleosome (1KX5): DNA in grey around histones. Zephyris, CC BY-SA 3.0.

**Like beads on a string. Prokaryotic DNA has no histones: it is "naked".**

# Beads on a string

A1.2.13




DNA helix → **Nucleosomes** → Coiled fibre → Loops → Metaphase chromosome

**Key idea**

Each level folds the one before, packing DNA tighter and tighter.

# Why package DNA?

A1.2.13

| Fact | Size |
| --- | --- |
| **Human DNA per nucleus** | About 2 m |
| **Nucleus** | About 5–10 μm |
| **Metaphase chromosome** | About 5 μm long |

1. Fits 2 m into the nucleus
2. Condensed chromosomes move in division without tangling
3. Unwinding regions controls which genes are read

**Link**
D2.2 (HL): histone changes control packing and gene expression.

# Task: explore a nucleosome in 3D

A1.2.13

**Open**

PDB 1KX5 in Mol (rcsb.org) or JSmol

**Find**

The two copies of each of the four histones

**Colour**

Lysine and arginine: where are they?

**Report**

Describe where the positive amino acids sit relative to the DNA, and explain why.

# Eukaryotic and prokaryotic DNA

A1.2.13

| | Eukaryotes | Prokaryotes |
|---|---|---|
| **Histones** | Yes: nucleosomes | No: "naked" DNA |
| **Shape** | Linear chromosomes | Circular chromosome (+ plasmids) |
| **Location** | Nucleus | Nucleoid in cytoplasm |

**Link**

A2.2: prokaryote cell structure; B2.3 / D1.2: gene expression.

# Practice: packing

A1.2.13

**Describe**

A nucleosome, with H1 and linker DNA.

**Explain**

Why histones bind any DNA sequence.

**Calculate**

2 m into 5 μm: how many times shorter?

**Use**

histone, linker, positive, negative, condensed

# Check-in: class 2

A1.2.13

1. Describe the structure of a nucleosome.
2. Explain why histones attract DNA.
3. Give two reasons for packing DNA.
4. Distinguish eukaryotic from prokaryotic DNA.
5. State the role of H1.

**How to use this**

Answer without your notes, then check against slides 11–16.

# Hershey and Chase, 1952

A1.2.14



Martha Chase and Alfred Hershey. CC BY-SA 4.0.

**Bacteriophages** are DNA in a protein coat. Protein has **sulfur** but almost no phosphorus; DNA has **phosphorus** but no sulfur.

So they labelled protein with **sulfur-35** and DNA with **phosphorus-32** and followed each.

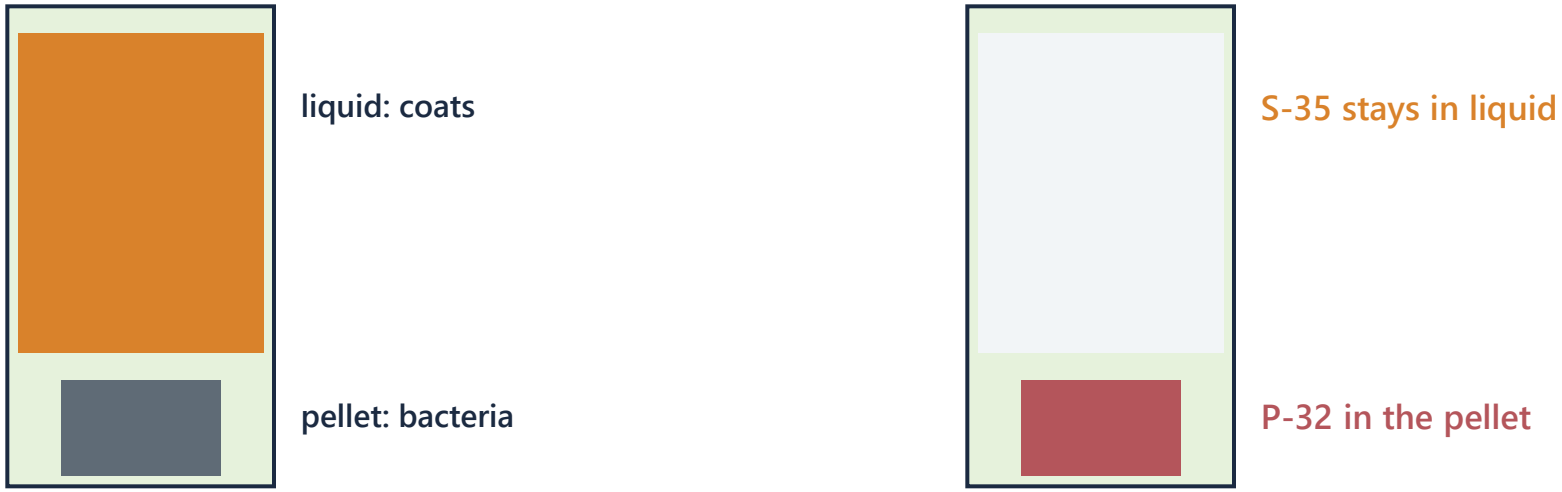
### Predict

If protein is the genetic material, which label should end up inside the bacteria? If DNA is?

# The experiment step by step

A1.2.14

Phages infect E. coli → Blender shakes coats off → **Centrifuge** → Measure radioactivity




Coloured = where most radioactivity was found. Schematic.

# What Hershey and Chase found

A1.2.14

| Result | S-35 (protein) | P-32 (DNA) |
|---|---|---|
| **In pellet (bacteria)** | Very little | Most |
| **In liquid (coats)** | Most | Little |
| **In new phages** | None | Some |

Only phage **DNA** entered the bacteria, yet they made complete new phages. So **DNA** carries the genetic information.

**Technology**
Radioactive isotopes became widely available after 1945.

# Evaluating the evidence



### Why protein was expected

20 amino acids seemed to offer far more variety than four bases.

### Not 100% clean

Some S-35 reached the pellet and some P-32 stayed in the liquid. The difference was still large and consistent.

### Discuss

Avery (1944) showed purified DNA could transform bacteria. Why is agreement between different methods convincing?

# Chargaff's data on base composition



| Source | A % | G % | C % | T % |
|---|---|---|---|---|
| **Human** | 30.9 | 19.9 | 19.8 | 29.4 |
| **Wheat** | 27.3 | 22.7 | 22.8 | 27.1 |
| **E. coli** | 24.7 | 26.0 | 25.7 | 23.6 |
| **Yeast** | 31.3 | 18.7 | 17.1 | 32.9 |

In every species **A = T** and **G = C**.

But A + T compared with G + C **varies between species**.

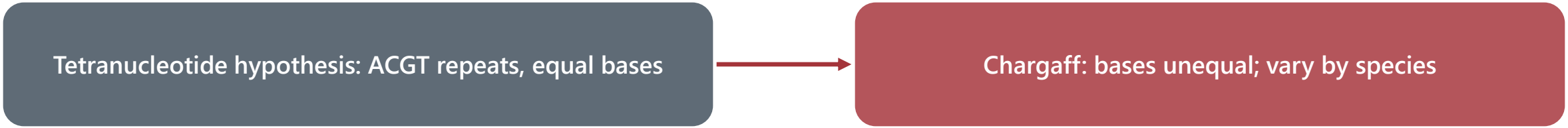
Chargaff and colleagues, about 1950 (rounded).

**Try it**

Calculate (A + G) / (C + T) for each species. Explain the result.

# How Chargaff's data falsified a hypothesis

A1.2.15




If DNA were a simple repeat it could carry no information. Chargaff's data **falsified** this idea.

His A = T, G = C rule was later explained by Watson and Crick's base pairing (1953).

### Nature of science

One clear contradiction can falsify a hypothesis; no number of agreeing results can prove one.

# Check-in: class 3

A1.2.14–15

1. Explain why sulfur and phosphorus were used as labels.
2. Explain why P-32 ended up inside the bacteria.
3. Give one limitation of the Hershey-Chase results.
4. State Chargaff's rules.
5. Explain how Chargaff falsified the tetranucleotide hypothesis.

**How to use this**

Answer without your notes, then check against slides 18–23.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **New nucleotides are added to the 5′ end.** | They are added to the 3′ end. |
| **Histones are made of DNA.** | Histones are proteins. |
| **Hershey and Chase showed the double helix.** | They showed DNA is the genetic material. |
| **Chargaff showed A = G and C = T.** | A = T and G = C. |
| **Prokaryotic DNA is on histones.** | It is naked. |

# Summary: HL nucleic acids

A1.2.11–15

| | Key idea |
|---|---|
| **A1.2.11** | 5′ and 3′ ends; antiparallel strands |
| **A1.2.12** | Purine + pyrimidine: constant width |
| **A1.2.13** | Nucleosomes: 8 histones, linker DNA, H1 |
| **A1.2.14** | Hershey-Chase: DNA enters, protein does not |
| **A1.2.15** | Chargaff: A = T, G = C; falsification |

# Check that you can...

1. **Explain** 5′ and 3′ ends and antiparallel strands
2. **Explain** why purine-pyrimidine pairing gives a regular helix
3. **Describe** a nucleosome
4. **Evaluate** the Hershey-Chase experiment
5. **Analyse** Chargaff's data
6. **Explain** falsification of the tetranucleotide hypothesis

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. State which end of a DNA strand new nucleotides are added to.
2. Describe the structure of a nucleosome.
3. Explain why the phosphorus-32 label ended up inside the bacteria in the Hershey-Chase experiment.

# Image credits

Nucleosome 1KX5 colour coded.png – Zephyris at English Wikipedia – CC BY-SA 3.0

Nucleotides 1.svg – Boris (PNG), SVG by Sjef – Public domain

Scientist Alfred Hershey and Martha Chase.jpg – Rohit Kumar Sengupta – CC BY-SA 4.0

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